

Functional Test - Module 5: Login

Project: Blue Team — Moffat Bay Marina

Course: CSD 460 - Capstone Project

Description: Test plan for the login popup, covering the signed-in header state, logout, the generic failure message, and account logout.

Date: 2026/09/05

Front End Developer: Miguel Fernandez

Back End Developer: Robert Breutzmann

Developer Tester: Miguel Fernandez

Peer Tester: Sara White

To complete the test case plan, fill out the information for Project, Course, Description and Date in the header above.

About this document

These tests were originally written and performed by Miguel Fernandez (developer) and Sara White (peer tester) for the Module 5 test plan (Module-5/BlueTeam_Mod5_TestPlan.docx , Tests 5, 6). They were re-formatted into this layout by Claude. The steps, results, dates, comments, and screenshots are carried over from the original unchanged; the original recorded each result as Yes or No, shown here as Pass or Fail. The tests were not re-run.

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For each test, the developer should: provide a test description, a test objective, the developer name and date tested. For each step, fill out the actions to be taken and describe the expected results, and check Pass or Fail. The peer tester should provide their name, date tested, check Pass or Fail for each step, and fill out the Screenshots list below the table with the matching screenshot number for each item.

Test 1: Login

Test Objective: Verifies that the site shows the signed-in state in the shared header on every page, that landing page controls change once a customer is signed in, and that logging out ends the session.

DEVELOPER: Miguel Fernandez | TESTED ON: Sept 6, 2026

PEER TESTER: Sara White | TESTED ON: Sept 6, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	In a browser, navigate to http://localhost:8081/marinawebsite/ or http://localhost:8080/marinawebsite/	The Moffat Bay Marina landing page should load. The button at the far right of the header should read "Log In" and there should be no welcome text next to it.	Pass	Pass

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 1 - Landing page while signed out. The header shows Log In.

Step	Action	Expected Results	Dev Test	Peer Test
2	Click the "Log In" button in the header.	A sign in popup should open over the page with an "Email address" field, a "Password" field, a "Sign in" button, and a "Register here" link. The page behind it should dim.	Pass	Pass

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 2 - After clicking Log In, a sign in popup appears.

Step	Action	Expected Results	Dev Test	Peer Test
3	Enter elena.marsh@example.com in the email field and the matching seeded password in the password field, then click "Sign in".	The popup should close and the browser should return to the landing page. The header should now read "Welcome, Elena M." with a "Log Out" button in place of "Log In".	Pass	Pass

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 3 - After signing in. The header now shows Welcome, Elena M. and Log Out.

Step	Action	Expected Results	Dev Test	Peer Test
4	Scroll down to the "Ready to Reserve Your Slip?" section near the bottom of the landing page.	The section should read "Check availability and book your spot today" with a "Book a Slip" button. It should no longer offer to create an account, because the visitor already has one.	Pass	Pass

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 4 - The reservation section now reads Book a Slip instead of Create an Account.

Step	Action	Expected Results	Dev Test	Peer Test
5	Click "About Us" in the header navigation.	The About Us page should load and its header should also show "Welcome, Elena M." and "Log Out". The signed-in state is shown by the shared header, so it appears on every page without each page checking the session itself.	Pass	Pass

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 5 - The About Us page shows the same signed-in header, without any page-specific code.

Step	Action	Expected Results	Dev Test	Peer Test
6	Click the "Log Out" button in the header.	The browser should land on the landing page at http://localhost:8081/marinawebsite/ and the header should read "Log In" again. Logout deliberately returns to the landing page rather than the previous page, since that page may itself require a signed-in session.	Pass	Not recorded

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 6 - After logging out. Back on the landing page with the header showing Log In.

Step	Action	Expected Results	Dev Test	Peer Test
7	Press the browser's Back button.	The previous page may briefly appear with the signed-in header still showing. This is the browser's cached copy of the page, not a live session.	Pass	Pass

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 7 - The browser Back button shows a cached copy of the signed-in page.

Step	Action	Expected Results	Dev Test	Peer Test
8	Press F5 to reload that same page.	The header should now read "Log In", confirming the session was actually invalidated on the server and the previous view was only a cached page.	Pass	Pass

Peer tester (Sara)

Developer (Miguel) - Test 5, Step 8 - After pressing F5 the same page shows Log In, confirming the session ended.

Comments

Developer (Miguel)
All eight steps passed. The signed-in state is handled entirely by the shared header include, so it appeared correctly on the landing page and the About Us page without either page containing its own session check. The display name arrives from the session already formatted as first name plus last initial, so no page has to build it.

Step 7 is worth flagging for the team rather than counted as a failure. Pressing Back after logout shows a cached copy of the signed-in page. The session really is gone, as step 8 proves, but on a shared computer the next person could still glimpse the previous user's page from history. Adding no-cache response headers to pages that require a login would close that gap, and it is worth considering before the site is handed over.

The landing page hero button also changes behaviour once signed in. It links straight to the reservations page instead of opening the sign in popup, which removes the loop where a signed-in customer was still being asked to log in.

Peer Tester (Sara)
All eight steps worked as described and matched the results of the previous tester. I agree with Miguel's comment above stating that step 7 is worth being flagged for the team to review. While the session was not active (and this was confirmed upon reloading the page), it would be preferable if no personal information was displayed.

Test 2: Login

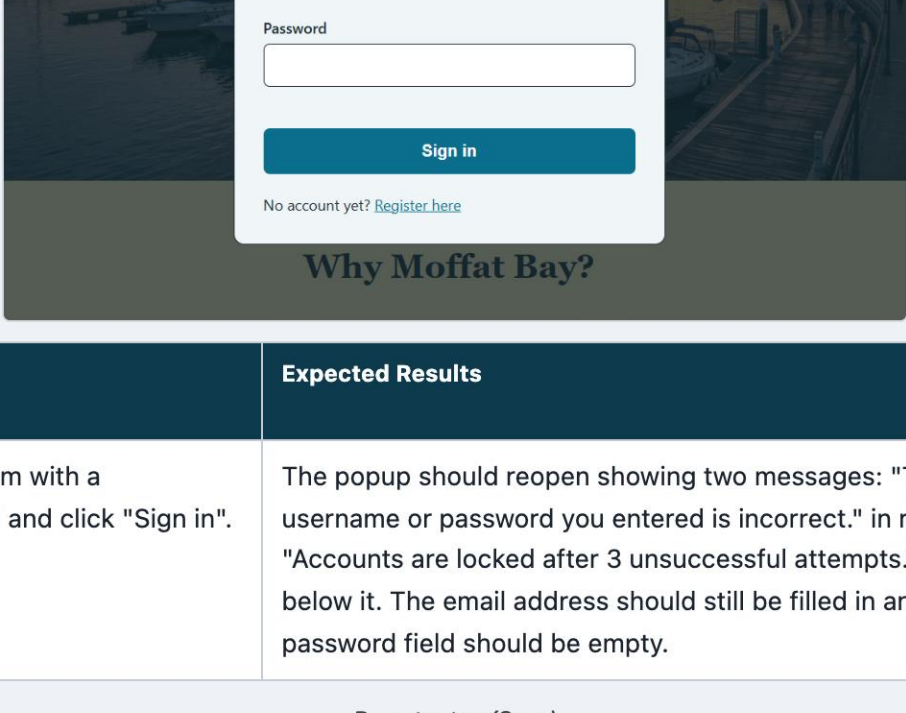
Test Objective: Verifies that a failed sign in shows the same generic message whether or not the email belongs to a real account, and that three consecutive failures lock the account until it is unlocked.

DEVELOPER: Miguel Fernandez | TESTED ON: Sept 6, 2026

PEER TESTER: Sara White | TESTED ON: Sept 6, 2026

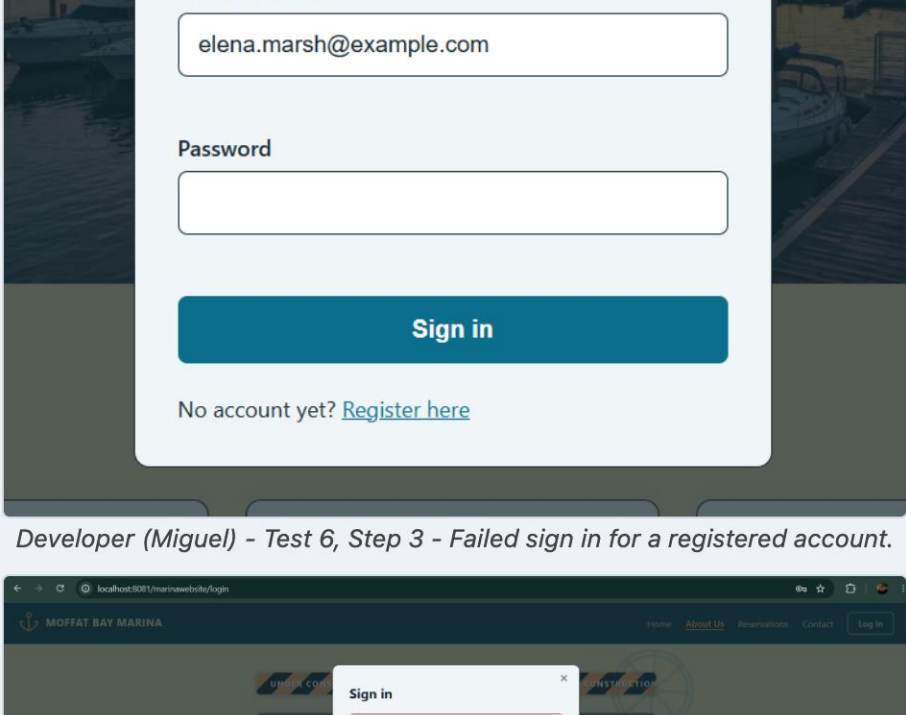
Step	Action	Expected Results	Dev Test	Peer Test
1	Reset the test account so the failure count starts at zero, using this MySQL statement: USE MoffatBayMarinaDB; UPDATE Customer SET failedLoginAttempts = 0, accountLocked = 0 WHERE email = 'elena.marsh@example.com';	The statement should report one row affected. The account is now unlocked with no recorded failures.	Pass	N/A* (see comments)

Peer tester (Sara)

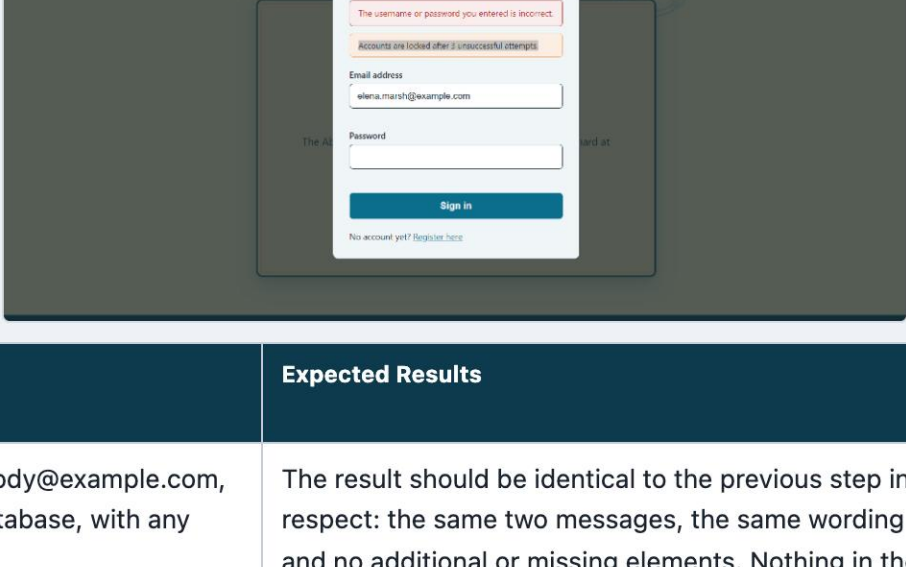


Step	Action	Expected Results	Dev Test	Peer Test
2	Navigate to http://localhost:8081/marinawebsite/ or http://localhost:8080/marinawebsite/ and click "Log In".	The sign in popup should open with empty email and password fields.	Pass	Pass

Peer tester (Sara)

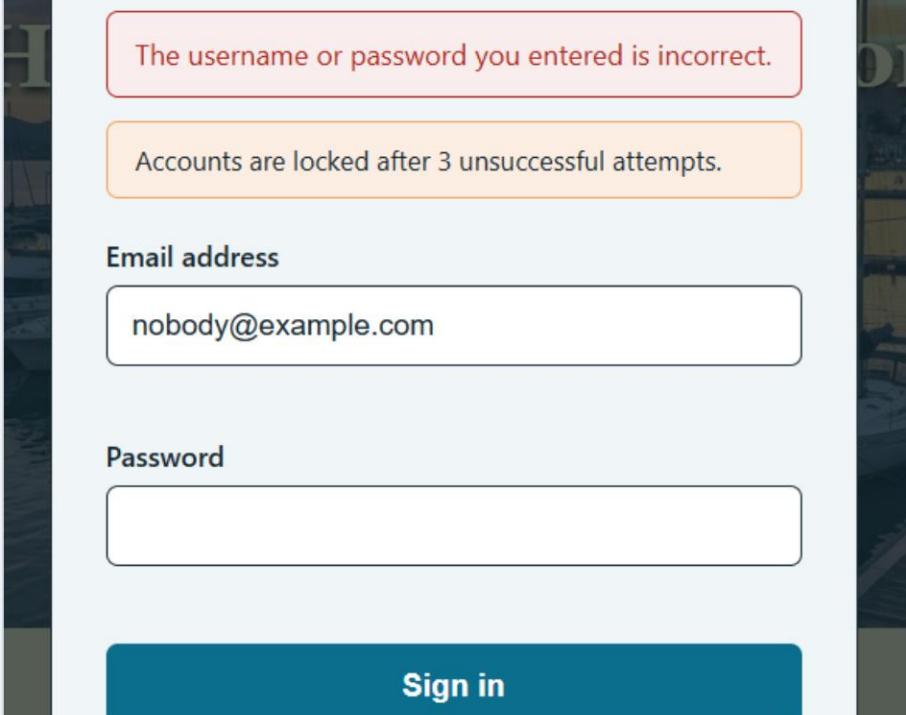


Developer (Miguel) - Test 6, Step 3 - Failed sign in for a registered account.

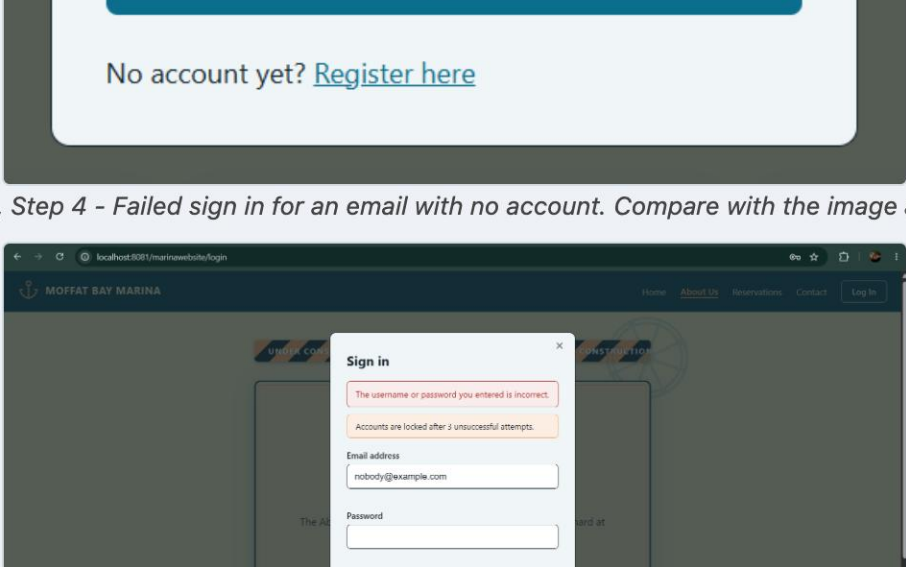


Step	Action	Expected Results	Dev Test	Peer Test
3	Enter elena.marsh@example.com with a deliberately incorrect password and click "Sign in".	The popup should reopen showing two messages: "The username or password you entered is incorrect." in red, and "Accounts are locked after 3 unsuccessful attempts." in amber below it. The email address should still be filled in and the password field should be empty.	Pass	Pass

Peer tester (Sara)

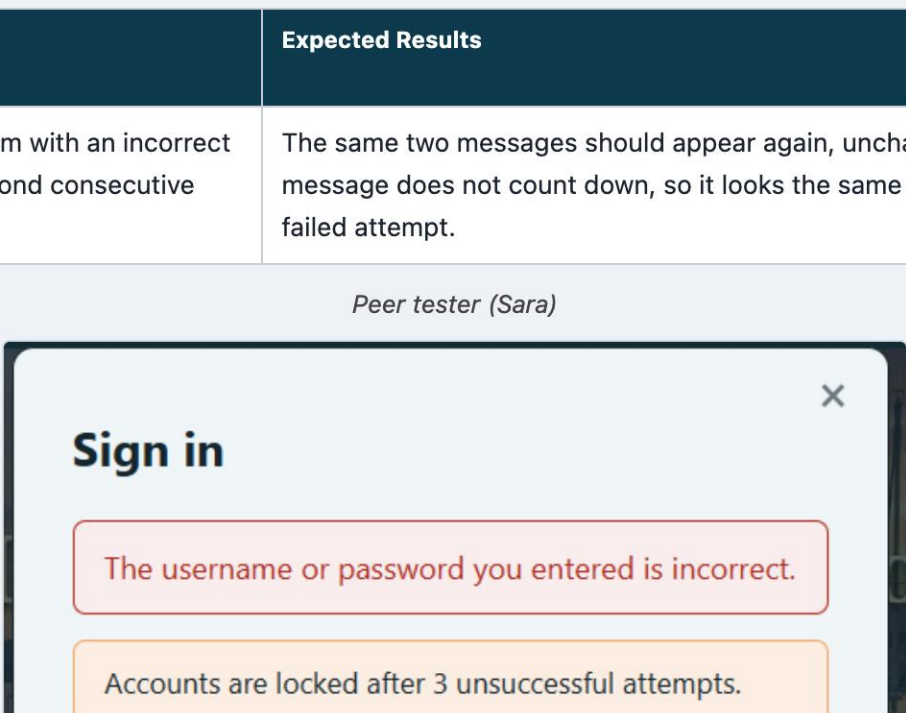


Developer (Miguel) - Test 6, Step 4 - Failed sign in for an email with no account. Compare with the image above: the two are identical.

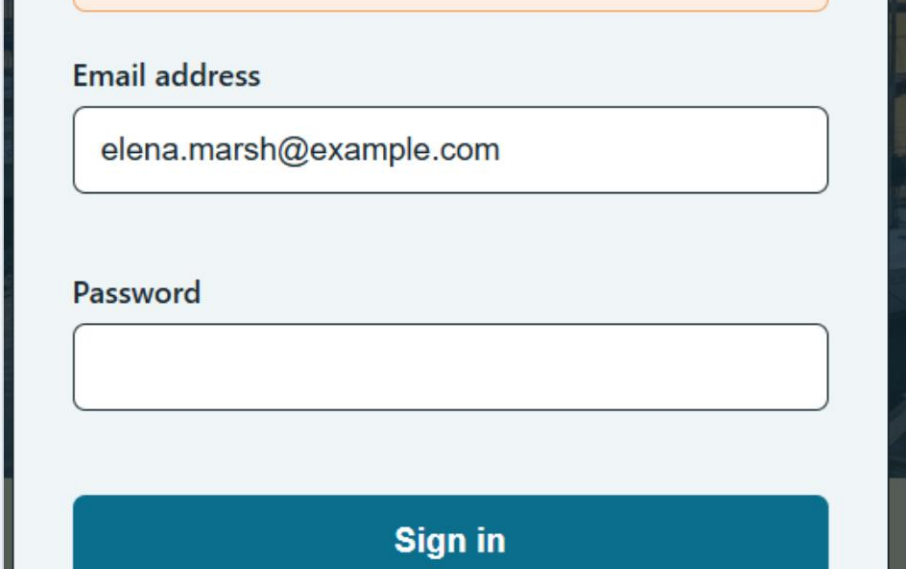


Step	Action	Expected Results	Dev Test	Peer Test
4	Clear the email field, enter nobody@example.com, which has no account in the database, with any password, and click "Sign in".	The result should be identical to the previous step in every respect: the same two messages, the same wording and styling, and no additional or missing elements. Nothing in the response should reveal that this email has no account.	Pass	Pass

Peer tester (Sara)

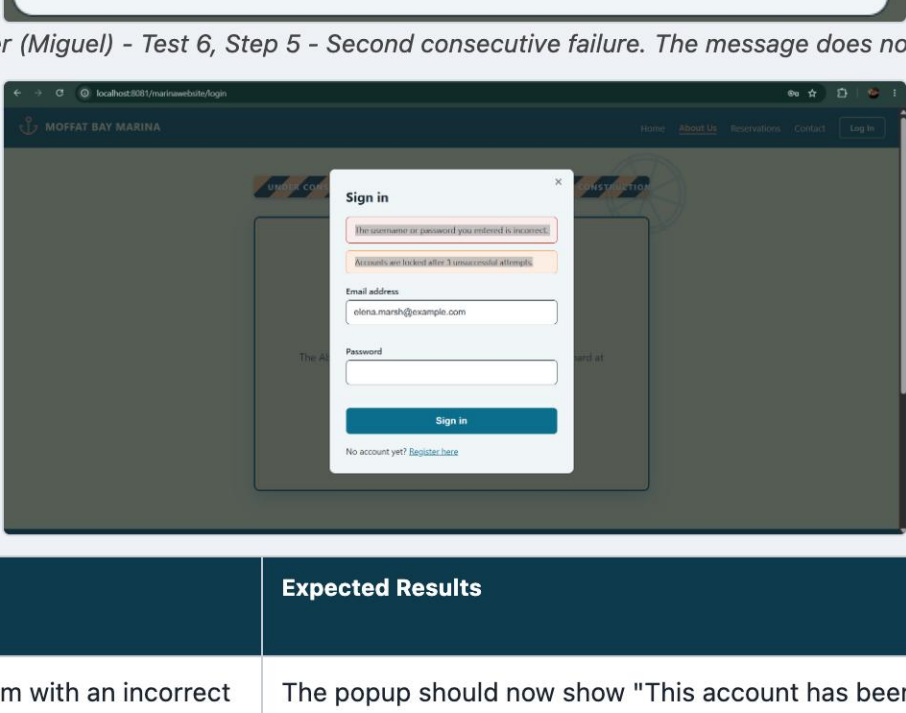


Developer (Miguel) - Test 6, Step 5 - Second consecutive failure. The message does not change.

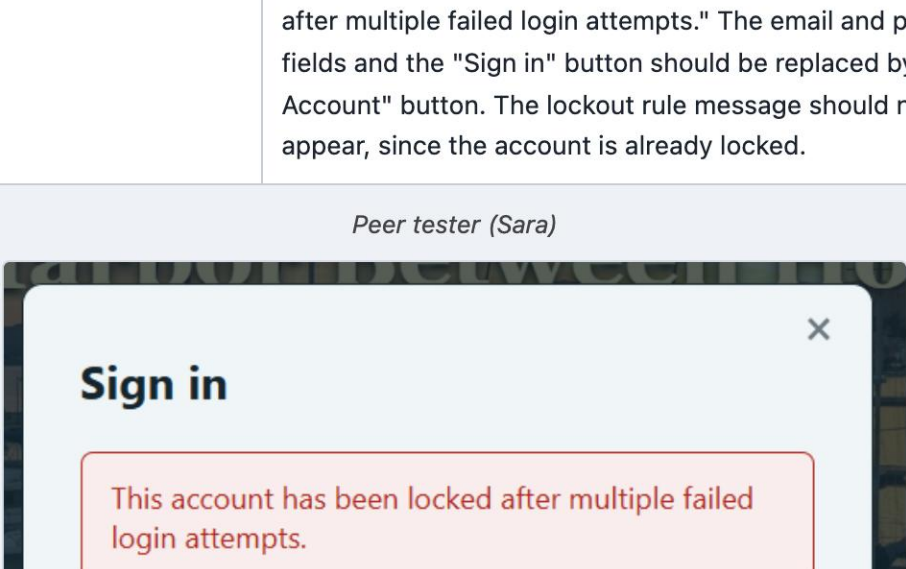


Step	Action	Expected Results	Dev Test	Peer Test
5	Enter elena.marsh@example.com with an incorrect password again. This is the second consecutive failure for that account.	The same two messages should appear again, unchanged. The message does not count down, so it looks the same on every failed attempt.	Pass	Pass

Peer tester (Sara)

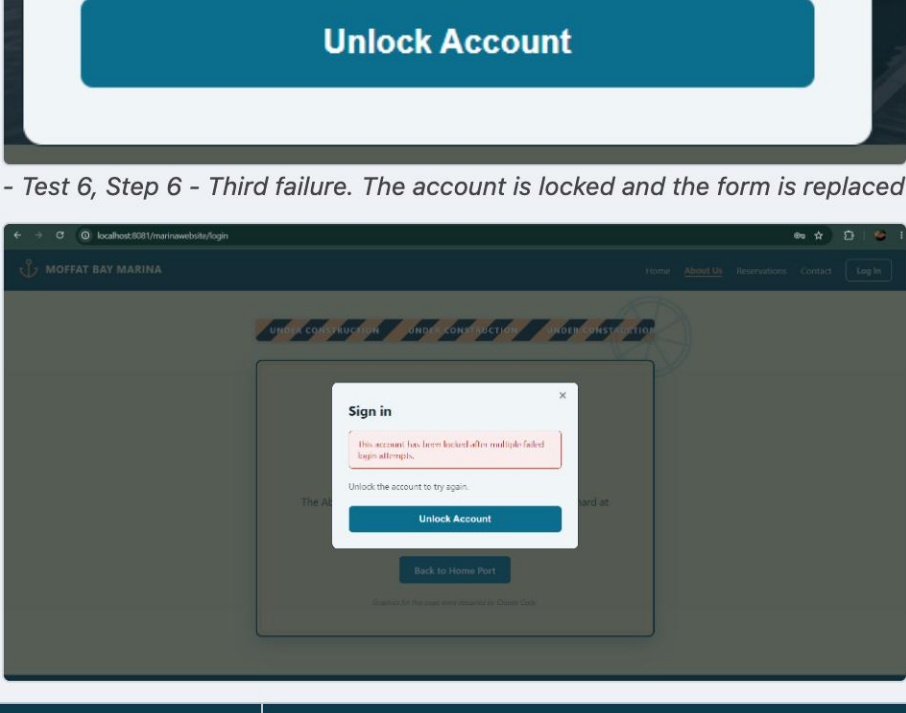


Developer (Miguel) - Test 6, Step 6 - Third failure. The account is locked and the form is replaced by 'Unlock Account'.

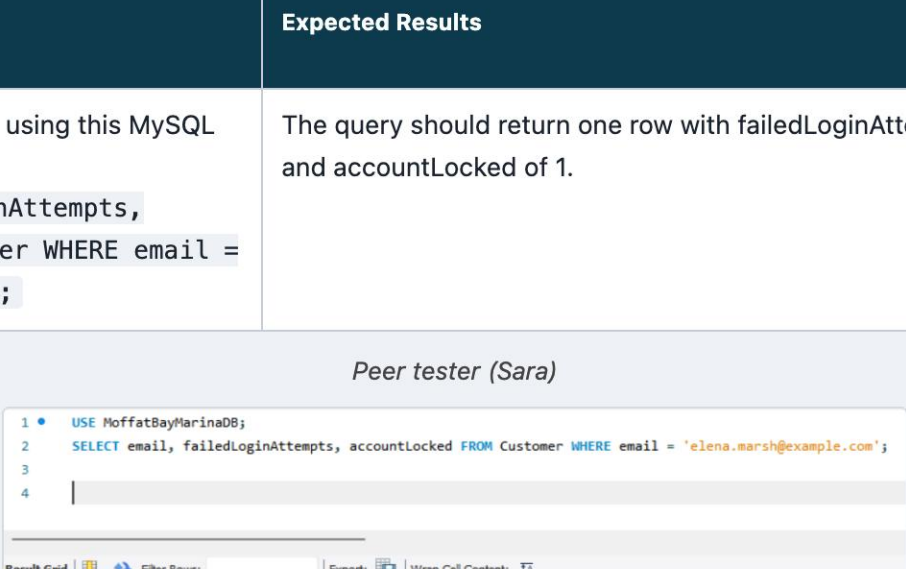


Step	Action	Expected Results	Dev Test	Peer Test
6	Enter elena.marsh@example.com with an incorrect password a third time.	The popup should now show "This account has been locked after multiple failed login attempts." The email and password fields and the "Sign in" button should be replaced by an "Unlock Account" button. The lockout rule message should no longer appear, since the account is already locked.	Pass	Pass

Peer tester (Sara)

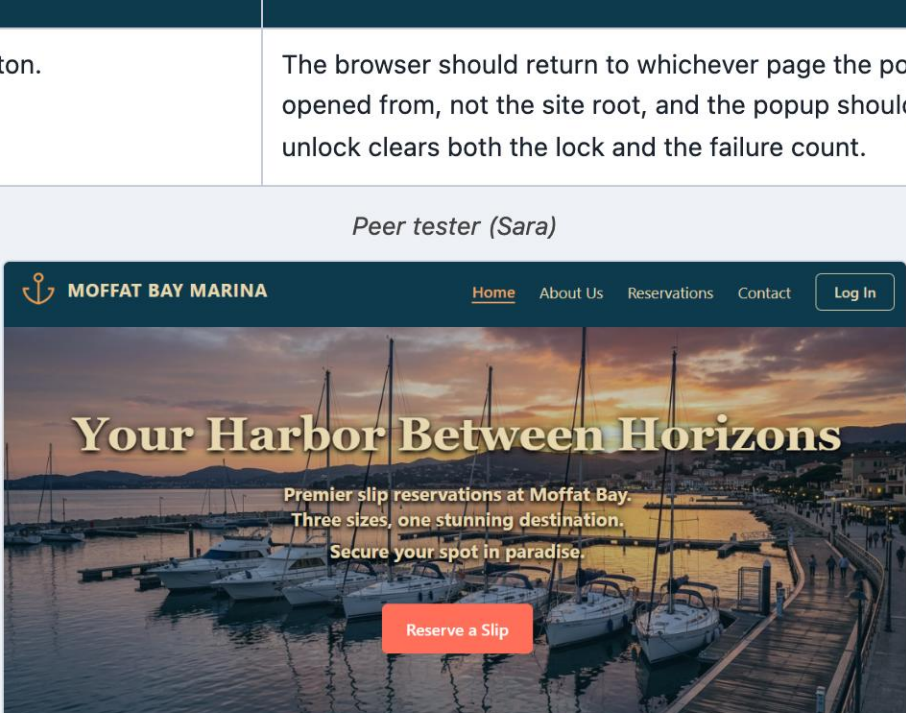


Developer (Miguel) - Test 6, Step 7 - Confirm the lock was recorded, using this MySQL statement:
SELECT email, failedLoginAttempts, accountLocked FROM Customer WHERE email = 'elena.marsh@example.com';

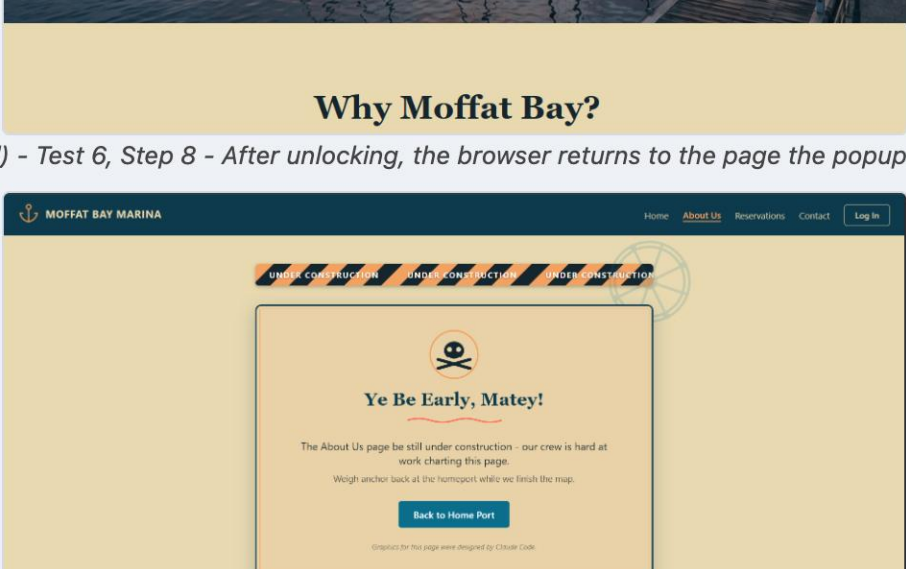


Step	Action	Expected Results	Dev Test	Peer Test
7	Confirm the lock was recorded, using this MySQL statement: SELECT email, failedLoginAttempts, accountLocked FROM Customer WHERE email = 'elena.marsh@example.com';	The query should return one row with failedLoginAttempts of 3 and accountLocked of 1.	Pass	Pass

Peer tester (Sara)

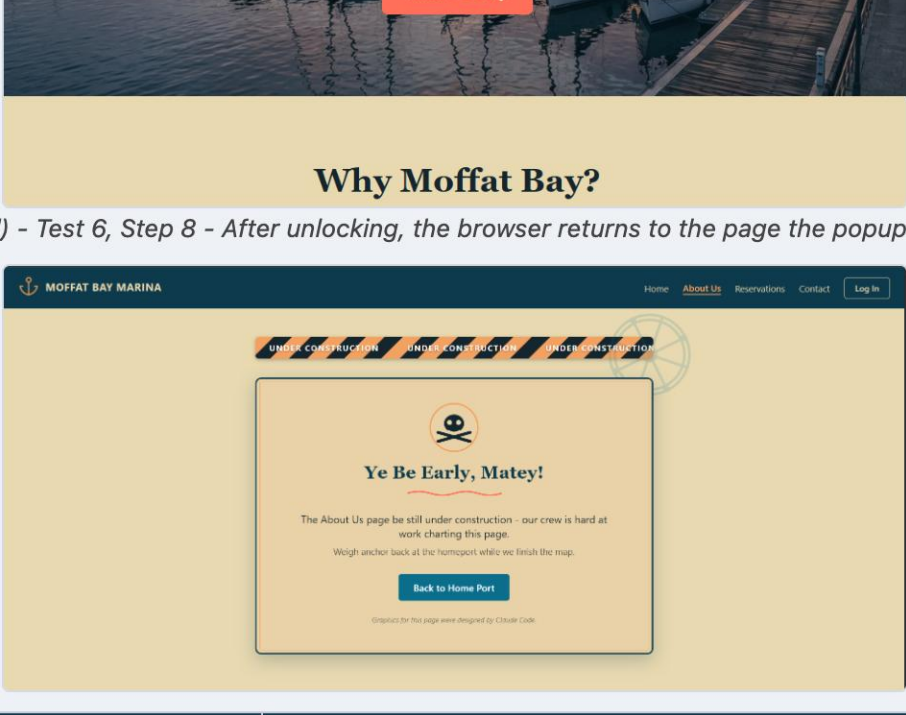


Developer (Miguel) - Test 6, Step 8 - After unlocking, the browser returns to the page the popup was opened from.

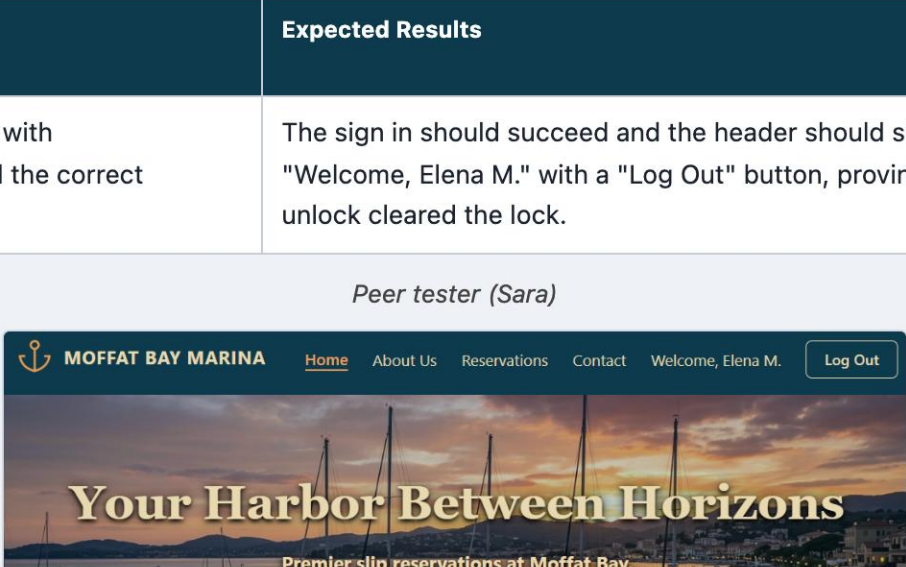


Step	Action	Expected Results	Dev Test	Peer Test
8	Click the "Unlock Account" button.	The browser should return to whichever page the popup was opened from, both the site root, and the popup should close. The unlock clears both the lock and the failure count.	Pass	Pass

Peer tester (Sara)

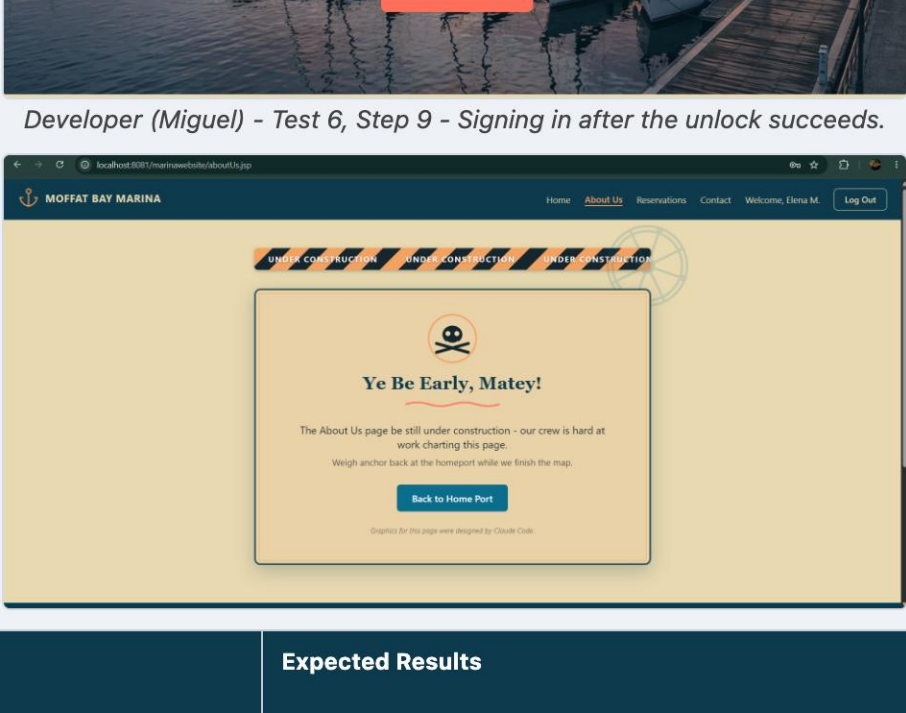


Developer (Miguel) - Test 6, Step 9 - Signing in after the unlock succeeds.

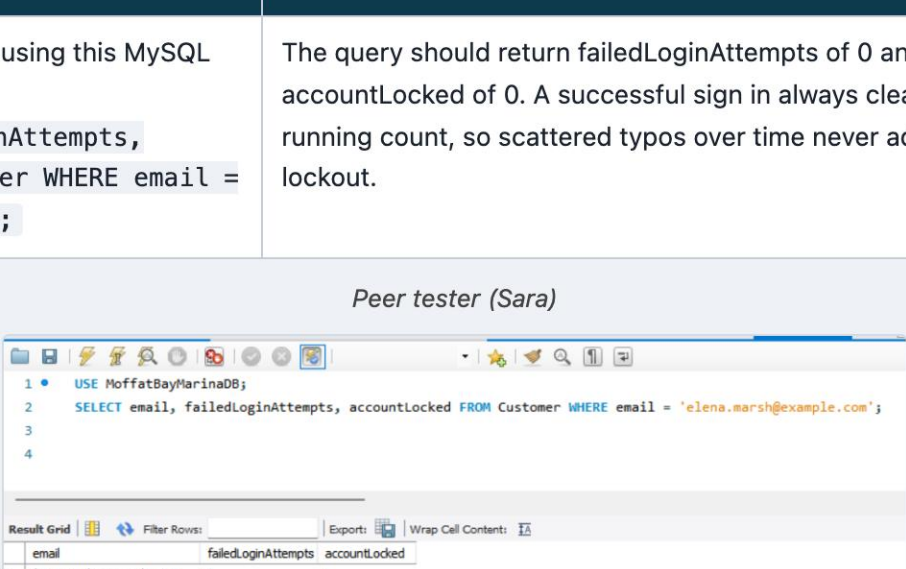


Step	Action	Expected Results	Dev Test	Peer Test
9	Click "Log In" again and sign in with elena.marsh@example.com and the correct password.	The sign in should succeed and the header should show "Welcome, Elena M." with a "Log Out" button, proving the unlock cleared the lock.	Pass	Pass

Peer tester (Sara)

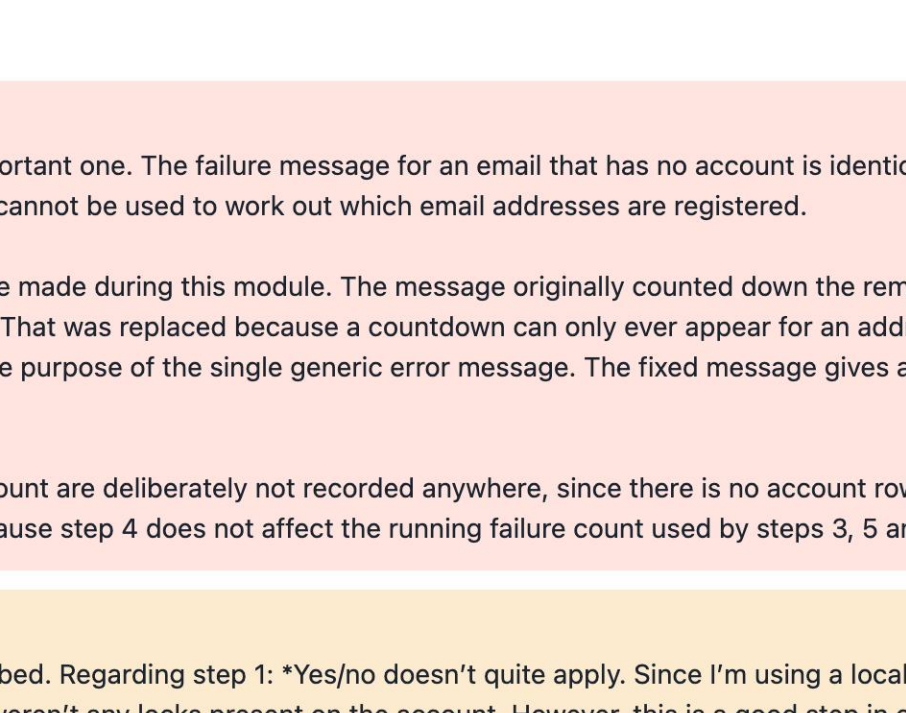


Developer (Miguel) - Test 6, Step 10 - Confirm the counter was reset, using this MySQL statement:
SELECT email, failedLoginAttempts, accountLocked FROM Customer WHERE email = 'elena.marsh@example.com';



Step	Action	Expected Results	Dev Test	Peer Test
10	Confirm the counter was reset, using this MySQL statement: SELECT email, failedLoginAttempts, accountLocked FROM Customer WHERE email = 'elena.marsh@example.com';	The query should return failedLoginAttempts of 0 and accountLocked of 0. A successful sign in always clears the running count, so scattered typos over time never add up to a lockout.	Pass	Pass

Peer tester (Sara)



Comments

Developer (Miguel)
All ten steps passed. Step 4 is the important one. The failure message for an email that has no account is identical to the message for a real account with a wrong password, so the popup cannot be used to work out which email addresses are registered.

This behaviour is the result of a change made during this module. The message originally counted down the remaining attempts, showing "2 more failed attempts will lock this account." That was replaced because a countdown can only ever appear for an address that exists, so watching for it confirmed valid accounts and undid the purpose of the single generic error message. The fixed message gives a real user the same useful warning while telling an attacker nothing.

Attempts against an email with no account are deliberately not recorded anywhere, since there is no account row to attach a count to. That is worth knowing when repeating this test, because step 4 does not affect the running failure count used by steps 3, 5 and 6.

Peer Tester (Sara)
All steps passed and worked as described. Regarding step 1: *Yes/no doesn't quite apply. Since I'm using a local copy of the database for the testing and development environment, there weren't any locks present on the account. However, this is a good step in case one might have existed.